

Kiran Kuruvila

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Professional Summary

Machine Learning Engineer focused on building AI systems, quantitative pipelines, and computer-vision based automation workflows. Experienced in transformers, deep learning architectures, agentic LLM systems, RAG and real-time data pipelines. Hands-on experience deploying containerized applications on AWS infrastructure using Docker, EC2, Nginx, Linux, and Amazon ECR.

Education

Marian College, Kuttikkanam, Kerala

2025 – 2027

Master of Computer Applications (MCA)

MG University, Kottayam, Kerala

2022 – 2025

Bachelor of Arts in Economics

Projects

Agentic Trading System | *Python, LangGraph, Pydantic*

- Designed a modular multi-agent workflow system using LangGraph for financial signal processing and decision orchestration
- Implemented typed state management and validation pipelines using Pydantic for deterministic execution reliability
- Built structured data ingestion and feature-engineering workflows for quantitative analysis pipelines

Quantaire: Stock Market Pattern Detection | *Python, YOLOv8, Playwright, NumPy, Docker, AWS ECR*

- Built a computer-vision pipeline for automated detection of technical stock-chart patterns using a custom YOLOv8 model
- Automated TradingView chart capture using Playwright with headless browser orchestration and retry-based session recovery
- Containerized inference services using Docker and deployed images through Amazon Elastic Container Registry (ECR)

GPT-Style Transformer for Algorithmic Reasoning | *PyTorch, NumPy*

- Built a decoder-only Transformer model for arithmetic sequence prediction tasks
- Designed synthetic datasets and preprocessing pipelines for supervised learning workflows
- Implemented evaluation and optimization workflows for training stability and performance analysis

NanoGPT: Character-Level Language Model | *PyTorch*

- Implemented a GPT-style architecture with causal self-attention and token embedding workflows
- Trained language models on structured text datasets using AdamW optimization and learning-rate scheduling

Gap Scanner Telegram Bot | *Python, yfinance, python-telegram-bot, Cron*

- Built an automated market gap scanner that fetches live stock data daily at 9:17 AM via a cron job and analyzes price gaps across a watchlist
- Implemented gap detection logic to flag stocks exceeding $\pm 1\%$ threshold, filtering actionable signals from noise
- Delivered structured gap reports to a private Telegram group using the Telegram Bot API, restricted to authorized members

Cloud Deployment & Backend Infrastructure | *Django, MySQL, AWS, Linux, Docker, Nginx*

- Deployed containerized applications on AWS EC2 instances with Nginx reverse proxy configuration
- Worked with Linux server environments, deployment pipelines, process management, and remote infrastructure administration
- Managed relational database workflows using MySQL and AWS RDS

Neural Network Foundations | *Python, PyTorch*

- Implemented deep learning architectures from first principles including autograd engines, RNNs, Transformers, and CNNs.

Teaching & Knowledge Sharing

Neural Networks Fundamentals Session

- Conducted peer-level sessions explaining neural network fundamentals from first principles
- Taught gradient descent, backpropagation, optimization dynamics, and neural computation using mathematical intuition

Technical Skills

Languages: Python, SQL

ML / AI: PyTorch, Transformers, CNNs, RNNs(GRUs,LSTM), Reinforcement Learning, LangGraph, RAG

Data & Backend: NumPy, Pandas, scikit-learn, Django

Cloud & DevOps: AWS (EC2, RDS, ECR, EFS), Docker, Linux, Nginx, Git

Concepts: Feature Engineering, Agent Workflows, Data Pipelines, Model Deployment

Mathematics: Probability, Statistics, Linear Algebra, Calculus